L&T Semiconductor

BUILDING A SMARTER TOMORROW

PROJECT NAME

REVERSE ENGINEERING OF CALCULATOR

Explore
Disassemble
Understand



- PCB
- IC & Components
- Power System
- Keypad
- LCD Display

MEMBER NAME

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From Components to Computation

INTRODUCTION OF PRODUCT



A Casio Scientific Calculator is an electronic calculating device designed to perform a wide range of mathematical, scientific, and engineering calculations. Unlike a basic calculator, a scientific calculator can perform advanced operations such as trigonometry, logarithms, powers, roots, fractions, statistics, equations, matrices, and complex-number calculations.

Casio scientific calculators are widely used by school and college students, engineers, scientists, and professionals because they provide accurate results quickly and reduce the time required for lengthy mathematical calculations.

- ***Purpose of the Product***
- *The main purpose of a scientific calculator is to make complex mathematical calculations faster, easier, and more accurate. It reduces manual calculation errors and helps students solve mathematical and engineering problems efficiently.*





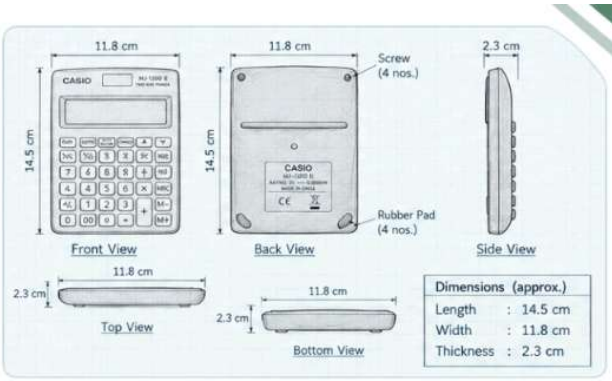
Physical Construct – Calculator

The selected product for reverse engineering is an old Casio MJ-120D II calculator. It is a compact, rectangular electronic device designed for basic mathematical calculations.



Approximate Dimensions

Length	Width	Thickness
14.5 cm	11.8 cm	2.3 cm



Dimensions (approx.)	
Length	: 14.5 cm
Width	: 11.8 cm
Thickness	: 2.3 cm

Physical Construction

Outer body
Rigid black plastic casing that protects the internal components.

Display
Rectangular LCD screen to display numbers and calculation results.

Keypad
Multiple plastic push buttons in rows and columns (numerical, arithmetic, memory, clear and function keys).

Rear casing
Back cover made of plastic and secured using screws, allowing access to internal components during servicing.

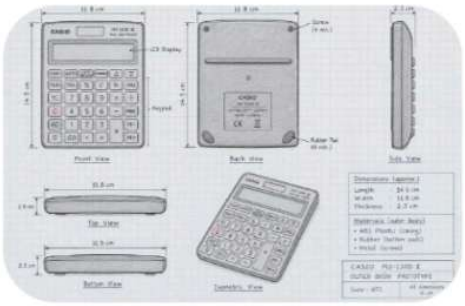
Base / support
Small rubber pads at the bottom to prevent slipping and protect the surface.

Power
Powered by an internal battery / solar-assisted power arrangement, depending on the model configuration.

Overall construction
Plastic enclosure, LCD display, membrane/ button-based keypad, screws, and supporting components assembled into a lightweight and portable structure.

Physical Construct – Calculator

The selected product for reverse engineering is an old Casio MJ-120D II calculator. It is a compact, rectangular electronic device designed for basic mathematical calculations.





Physical construction:

- **Outer body:** The calculator has a rigid **black plastic casing** that protects the internal components.
- **Front panel:** The upper portion contains an **LCD display**, while the lower portion contains the keypad.
- **Display:** A rectangular **LCD screen** is provided to display numbers and calculation results.
- **Keypad:** It consists of multiple **plastic push buttons** arranged in rows and columns. The keys include numerical keys, arithmetic operators, memory keys, clear keys, and function keys.
- **Rear casing:** The back cover is made of plastic and is secured using **screws**, allowing access to the internal components during servicing.
- **Base/support:** Small **rubber pads** are provided at the bottom to prevent slipping and protect the surface.
- **Power:** The calculator is powered using an internal battery/solar-assisted power arrangement, depending on the model configuration.
- **Overall construction:** The product uses a **plastic enclosure, LCD display, membrane/button-based keypad, screws, and supporting components** assembled into a lightweight and portable structure.

Subsystems of a Calculator
(Description)

1. Power Supply Subsystem

Provides the required electrical power to all the components of the calculator.



Coin cell battery (B+/B–)



Solar panel

Components:
- Coin cell battery
- Solar panel
- Diodes, resistors, capacitors (power management)

Function:
Supplies stable DC voltage (usually 3V) to the circuit and charges/supports the battery using solar energy.

2. Main Control Subsystem

Processes the input, performs calculations and controls the display.



Integrated Circuit (IC)

Components:
- Microcontroller / IC
- Resistors, capacitors
- Crystal (for timing)

Function:
- Reads key inputs
- Performs arithmetic operations
- Sends data to the LCD



3. Input Subsystem (Keypad)

Detects the key presses and sends the information to the main control unit.



Rubber keypad

Components:
- Rubber keypad
- Copper contacts on PCB

Function:
- Each key press closes a circuit on the PCB
- The controller identifies which key is pressed

4. Display Subsystem

Shows the numbers, symbols and results of calculations.



Components:
- LCD (Liquid Crystal Display)
- Connector / flexible cable

Function:
- Receives signals from the controller
- Displays the input, operations and result

5. User Interface Subsystem

Allows the user to reset the calculator and use the basic functions.



Reset button



Reset switch (on PCB)

Components:
- Reset button
- Reset switch (PCB)

Function:
- Clears the memory and restarts the system

6. Connection Subsystem

Connects all the parts together and provides proper signal paths.



PCB to LCD connector

Components:
- PCB tracks (copper lines)
- Connectors / flex cable
- Solder joints

Function:
Carries power, data and control signals between subsystems (PCB, LCD, keypad, etc.)

7. Processing Subsystem (Arithmetic Logic)

Performs the actual mathematical operations like addition, subtraction, multiplication and division.



Components:
- Main IC / Microcontroller

Function:
- Executes arithmetic logic
- Controls the flow of data to the display and keypad

MODES OF OPERATION – CASIO MJ-120D

Reverse Engineering Activity • Functional operations

1. Normal Calculation

Addition, subtraction, multiplication and division using +, −, ×, ÷ and =.

2. Percentage & Mathematical

%, √ and +/− provide percentage, square-root and sign-change functions.

3. Commercial Calculation

TAX+, TAX− and MU support tax and mark-up calculations.

4. Memory Operation

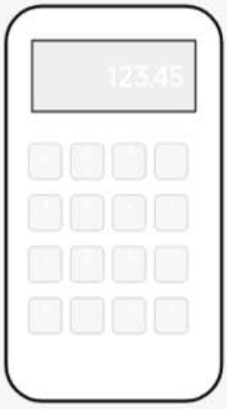
M+, M− and MRC store, modify and recall numerical values.

5. Review & Correction

150 Steps Check, AUTO REVIEW, GO TO and CORRECT help review and correct calculations.

6. Clear / Reset

C and AC clear entries or reset the current calculation.



CASIO MJ-120D

PURPOSE OF THE CALCULATOR

The Casio MJ-120D is designed to make everyday numerical calculations faster, easier and more reliable.

01 **Fast Arithmetic**
Performs addition, subtraction, multiplication and division quickly.

02 **Percentage & Mathematics**
Supports %, square-root and sign-change calculations.

03 **Commercial Calculations**
TAX+, TAX− and MU assist with tax and mark-up calculations.

04 **Memory & Review**
M+, M−, MRC and 150 Steps Check help store, review and correct calculations.

05 **Clear & Readable Results**
A 12-digit LCD provides clear numerical output for easy reading.

06 **Everyday Convenience**
Useful for students, homes, shops and offices where quick calculations are needed.

PURPOSE → QUICK • ACCURATE • SIMPLE • VERSATILE CALCULATIONS

USER NEEDS ADDRESSED

CASIO MJ-120D | PART I – STUDYING THE PRODUCT

01 **Fast & Accurate**

Quick arithmetic operations save time and reduce manual calculation errors.

02 **Mathematical Functions**

%, √ and +/− support percentage, square-root and sign-change calculations.

03 **Commercial Use**

TAX+, TAX− and MU simplify common tax and mark-up calculations.

04 **Memory & Review**

M+, M−, MRC and 150 Steps Check help store, review and correct calculations.

05 **Easy Operation**

Clearly arranged keys and a 12-digit LCD make the calculator easy to operate and read.

06 **Reliable & Convenient**

Compact construction with solar-cell assistance and an LR44 battery supports daily use.

KEY USER REQUIREMENT

Quick • Accurate • Easy to use • Versatile • Reliable

TECHNOLOGY COMPARISON • PHASE 1

Early Stage vs. First Generation

Early Stage (7-Segment LED/LCD)

- **Single-line display:** Strictly numerical digit outputs.
- **Dedicated ICs:** Fixed single-function microcontrollers.
- **Strict syntax:** Required rigid entry rules with no memory history.

1st Generation (S-V.P.A.M. Era)

- **Two-line dot matrix:** Expression on top, result below.
- **Multi-replay buffer:** Scroll back to correct entry errors.
- **Linear syntax:** Forced horizontal bracketed formula layout.



TECHNOLOGY COMPARISON • PHASE 2

Second Generation vs. Latest ClassWiz



2nd Generation (Natural-V.P.A.M.)

- **Textbook rendering:** Natural display for fractions and roots.
- **8-bit Microcontroller:** Handles real-time formula rendering.
- **Added solvers:** Equation solving and prime factorization.

Latest Generation (ClassWiz EX / CW)

- **High-res LCD screen:** 4x higher resolution pixel density.
- **32-bit ASIC Engine:** Rapid calculations with spreadsheets.
- **QR Code Connectivity:** Offloads graphs to mobile screens.

HARDWARE ARCHITECTURE

Current Status: Core Smart Technologies

High-Res Dot LCD

High pixel density enables icon menus and clear textbook math notation.

Dual Power Engine

Solar cells paired with low-drain CMOS logic extend battery past 3 years.

QR Cloud Link

Generates 2D QR codes to display dynamic plots on smartphone browsers.

High-Density ASIC

Embedded RAM supports 4×4 matrix ops and 30-item spreadsheets.

VALUE & IMPACT

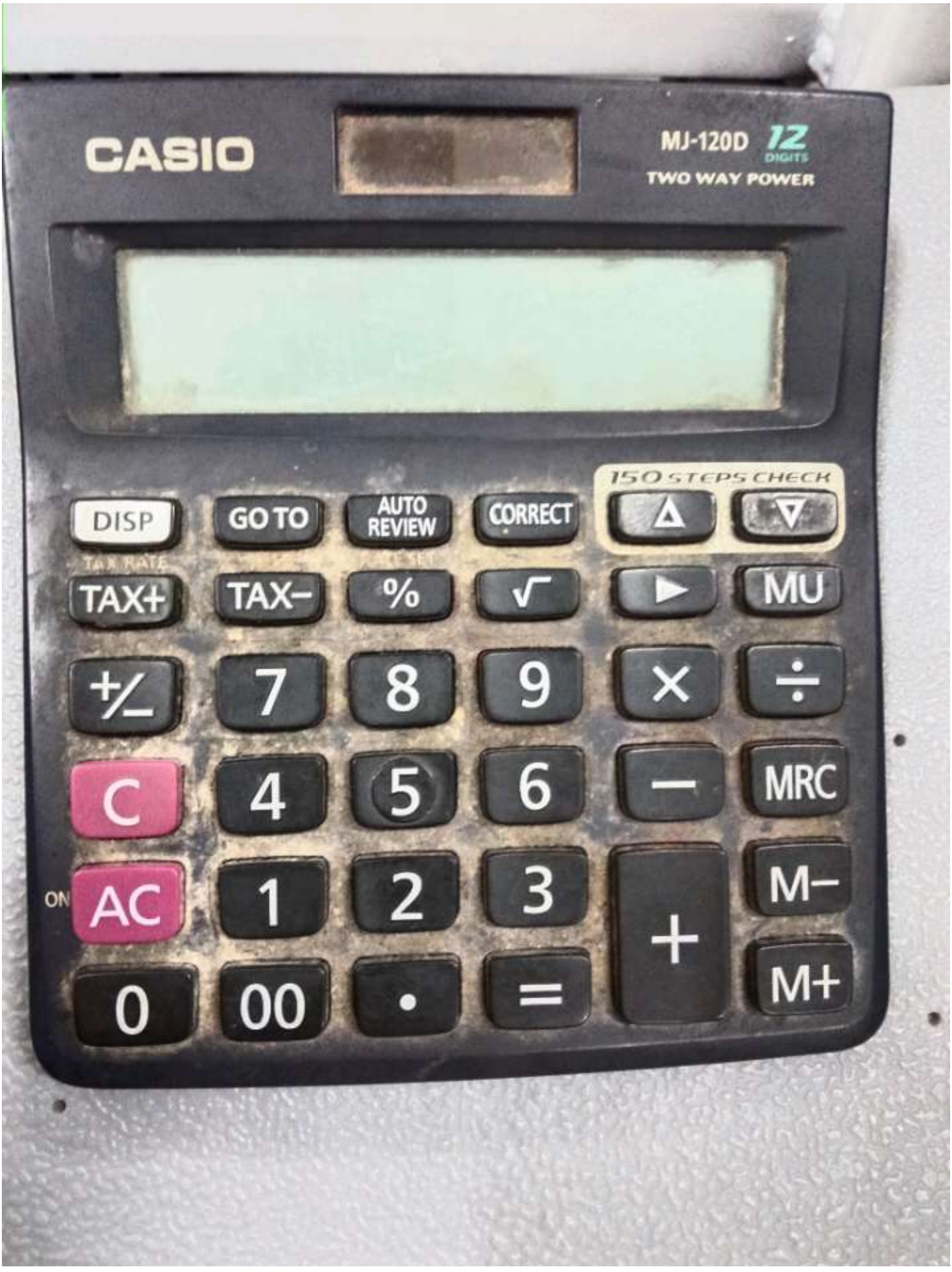
Engineering Importance of Evolution

1. **HCI Optimization**
   Natural notation drastically reduces user syntax mistakes during fast calculations.

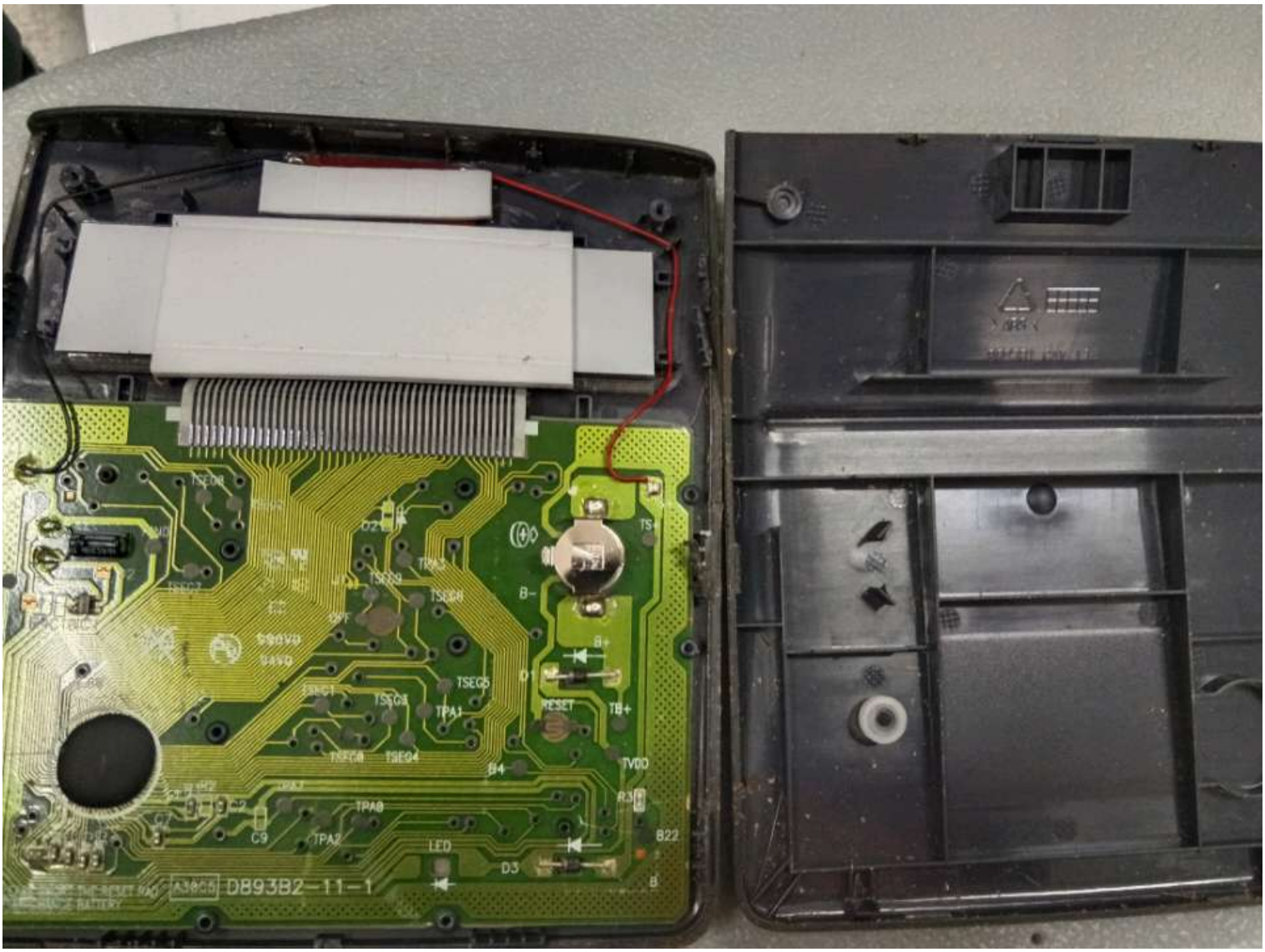
2. **Power Efficiency Architecture**
   Low operational voltage ensures uninterrupted primary solar power usage.

3. **Hardware-Software Convergence**
   QR bridging offloads high-cost 2D/3D plotting without adding battery drain.

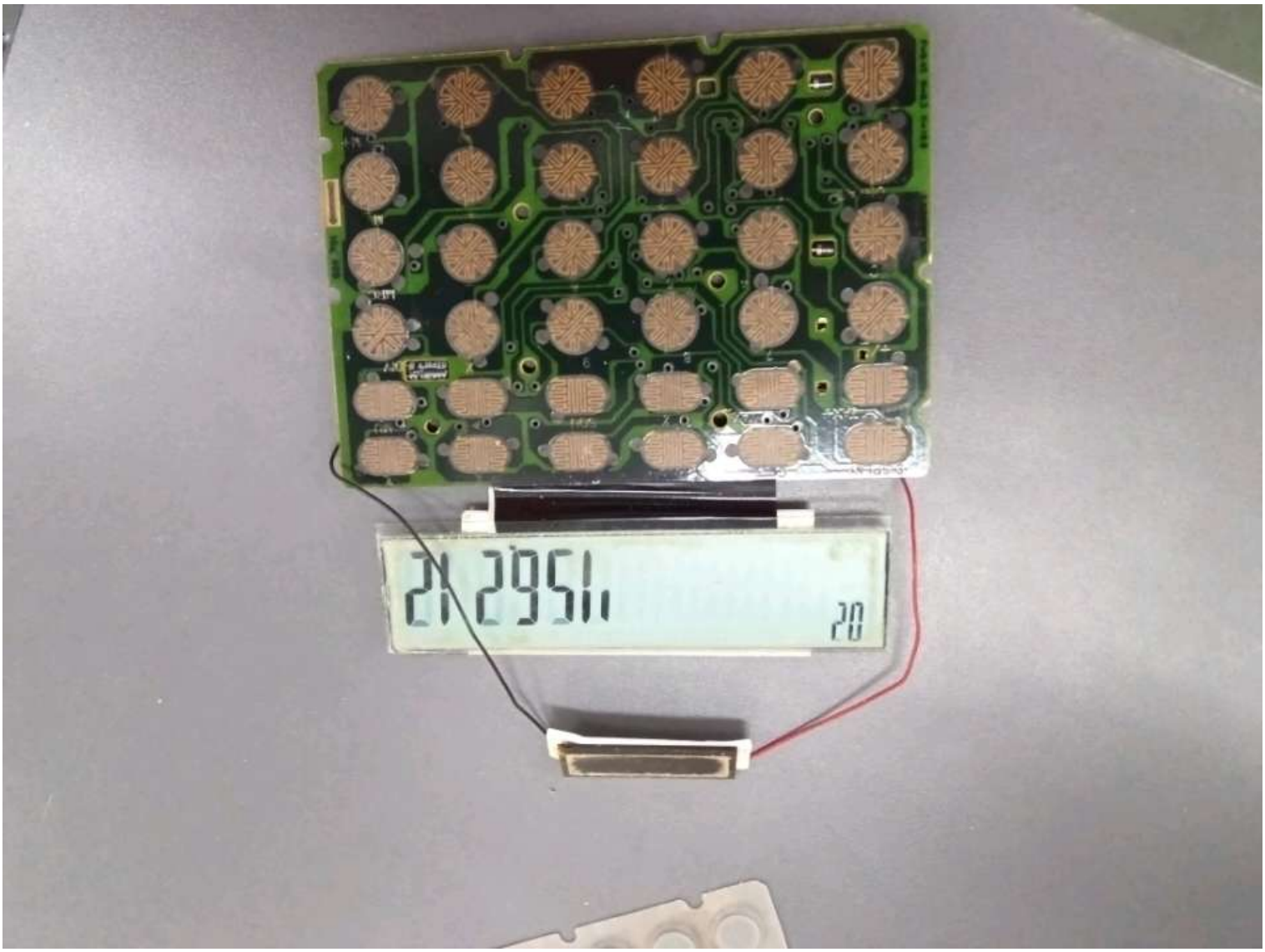
4. **Standardized Exam Reliability**
   Guarantees high-speed computation with zero network vulnerability.


CASIO
MJ-120D
12
DIGITS
TWO WAY POWER
150 STEPS CHECK
DISP
GO TO
AUTO REVIEW
CORRECT
TAX RATE
TAX+
TAX−
%
√
MU
+/−
7
8
9
×
÷
C
4
5
6
−
MRC
ON
AC
1
2
3
+
M−
0
00
•
=
M+


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CASIO


TSEG8
TPA3
TSEG9
TSEG8
B-
TS+
B+
TSEG5
TPA1
RESET
TB+
TSEG4
TVDD
B4
R3
TPA0
C9
TPA2
LED
D3
B22
D893B2-11-1


02
15:02 12

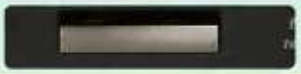

21 29511
20


C8
GND
TS-
J3
R4C10IC1
J2
TSEG7
E2
990V0
94V0
OFF
TSEG9
TSEG8
TPA3
D2
S+
TS+
B-
B+
D1
TSEG5
TSEG1
TSEG3
TPA1
RESET
TB+
TSEG0
TSEG4
B4
TVDD
R3
B22
TPA7
TPA0
R1
R2
C2
C7
C9
TPA2
LED
D3
C1
A30C5
D893B2-11-1
PLEASE SHORT THE RESET PAD
AFTER CHANGE BATTERY


9
AUTO REVIEW
X
MRC
MU

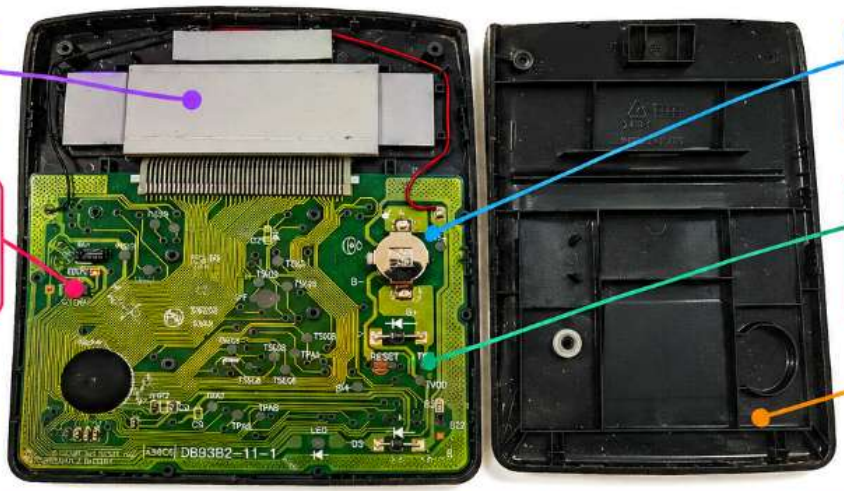

CASIO



Part Name	Image (from your photo)	Function
Screw		• Holds the calculator case together (keeps the back cover and front panel fixed).
Battery Compartment (cover area)		• Holds the button cell battery (provides power to the calculator when there is no light).
Solar Panel		• Charges the battery using light (extends battery life).
LCD Display		• Shows the numbers and calculations.
Keypad		• Buttons you press to enter numbers and operations. Each key has a rubber contact underneath.
Rubber Feet		• Provides grip and prevents slipping on surfaces.
Back Cover		• Protects the internal components and gives structure to the calculator.

LCD Display
Shows the numbers and calculations.

Circuit Board (PCB)
Houses the electronic components and connects everything.



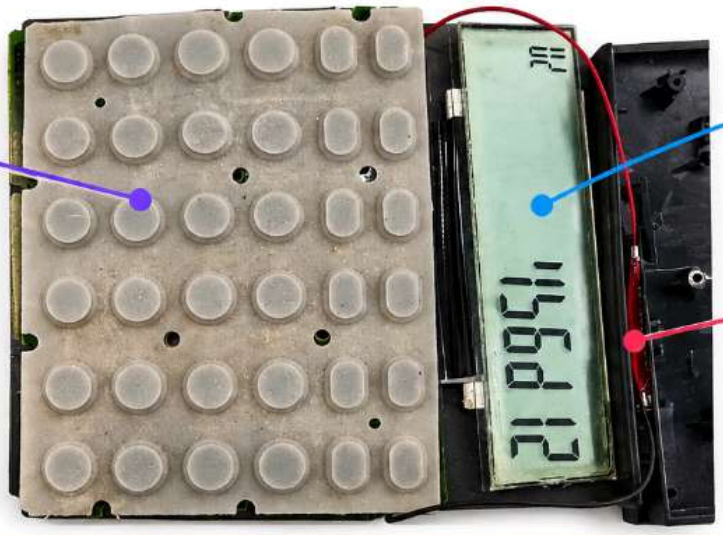
Battery (under the cover)
Provides power to the calculator when there is no light.

Reset Button
Resets the calculator to its default state (usually clears errors or memory).

Back Cover
Protects the internal components and gives structure to the calculator.

Part Name	Image (from your photo)	Function
LCD Display		• Shows the numbers and calculations.
Circuit Board (PCB)		• Houses the electronic components and connects everything in the calculator.
Battery (under the cover)		• Provides power to the calculator when there is no light.
Reset Button		• Resets the calculator to its default state (usually clears errors or memory).
Back Cover		• Protects the internal components and gives structure to the calculator.

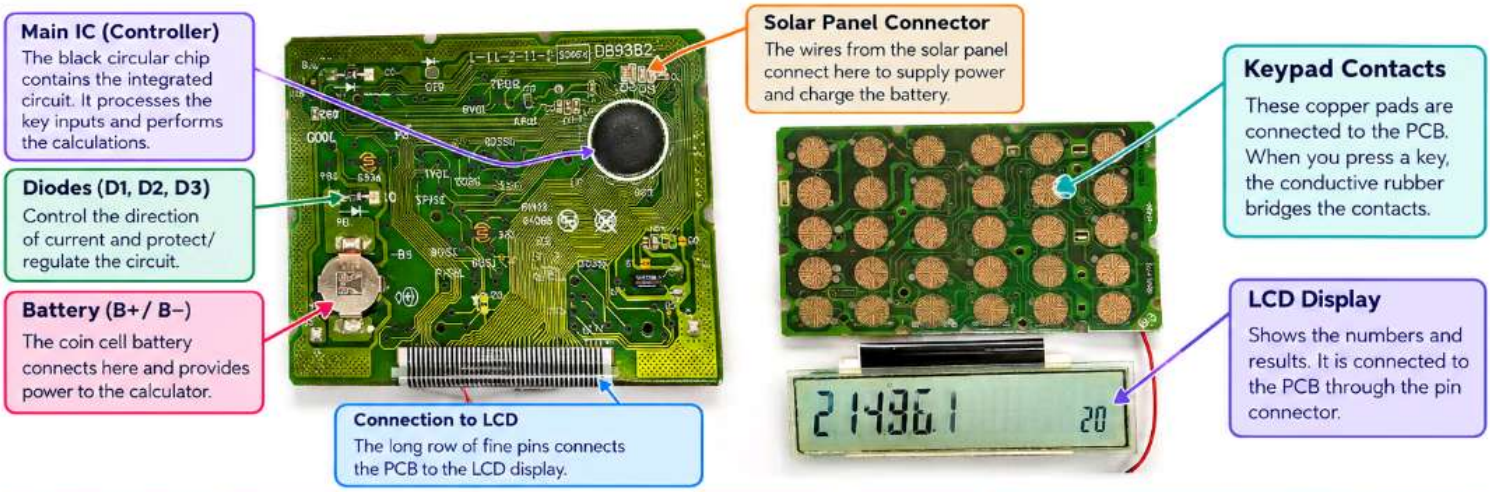
Rubber Keypad
These rubber buttons are pressed by the user. Each button has a conductive pad underneath.



LCD Display
Shows the numbers, operations and results on the screen.

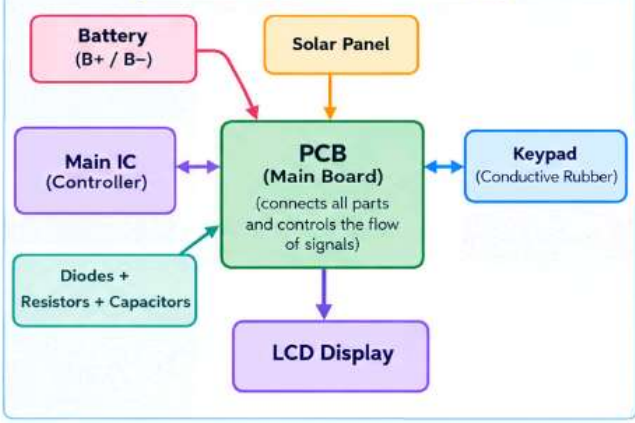
LCD Connector/Wires
Connects the LCD display to the main circuit board (PCB). Supplies the required voltage and signals.

Part Name	Image (from your photo)	Function
Rubber Keypad (Membrane Switch)		• Provides tactile response when a button is pressed. • Contains conductive pads that make contact with the PCB. • Each button corresponds to a specific function (number, +, −, ×, ÷, etc.).
LCD Display (Liquid Crystal Display)		• Shows the numbers, operations and calculations. • Uses liquid crystals that change on receiving electrical signals from the PCB.
LCD Connector / Wires		• Carries power (voltage) to the LCD. • Sends signals from the main circuit board so the display shows the correct information.

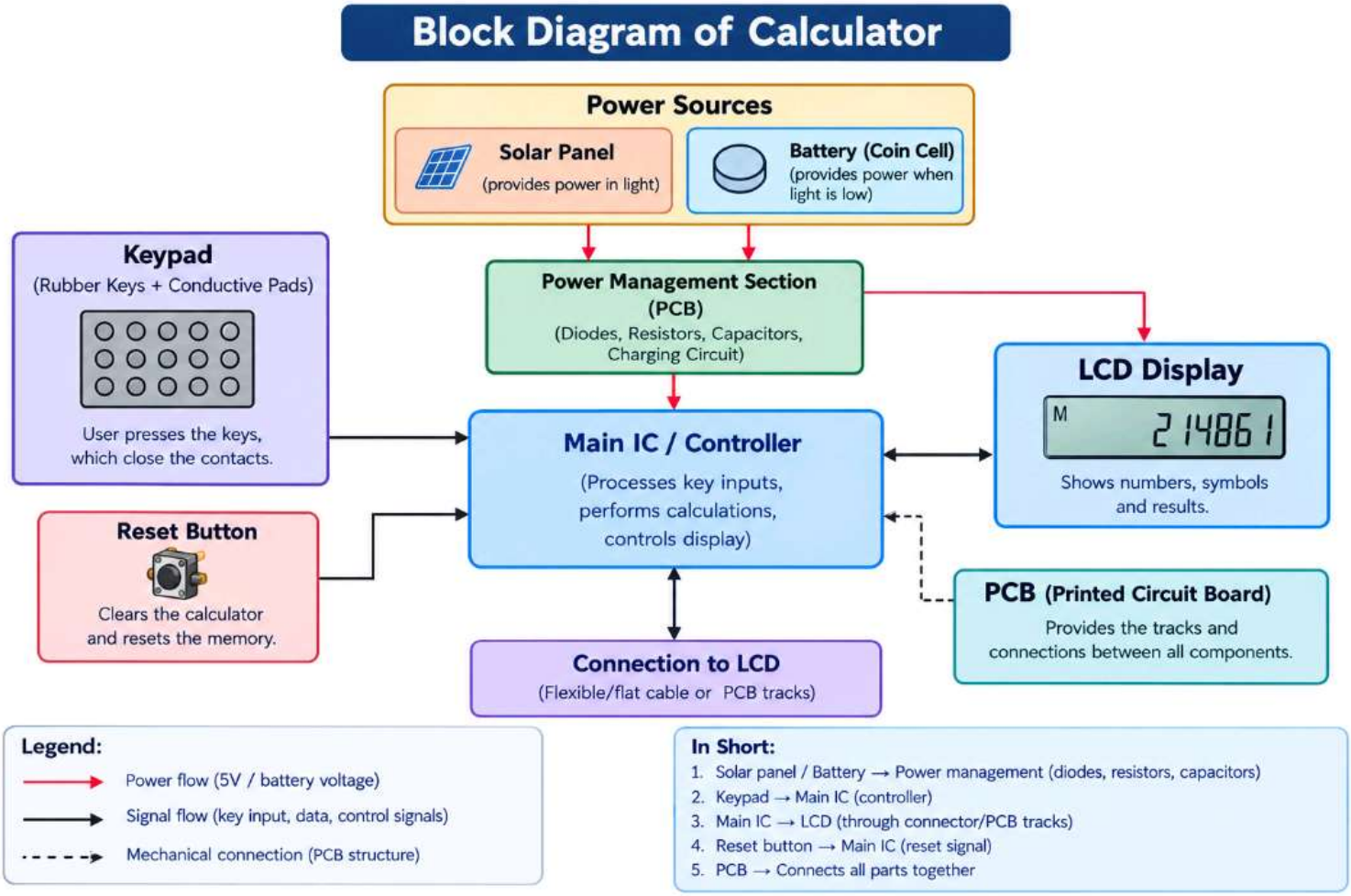


Part Name	Image (from your photo)	Function
Main IC (Controller)		• Processes key inputs and performs calculations.
Diodes (D1, D2, D3)		• Control current direction and protect the circuit.
Battery (B+/ B–)		• Provides power to the calculator.
Solar Panel Connector		• Supplies power and charges the battery (in light).
Connection to LCD		• Sends signals from the PCB to the LCD.
Keypad Contacts		• Detects key presses when the conductive rubber bridges the contacts.
LCD Display		• Shows numbers and results.

How They Are Connected



Block Diagram of Calculator



Legend:

- Power flow (5V / battery voltage)
- Signal flow (key input, data, control signals)
- Mechanical connection (PCB structure)

In Short:

1. Solar panel / Battery → Power management (diodes, resistors, capacitors)
2. Keypad → Main IC (controller)
3. Main IC → LCD (through connector/PCB tracks)
4. Reset button → Main IC (reset signal)
5. PCB → Connects all parts together

CASIO MJ-120D (12 Digit)

Component Rating & Specification

(Based on the Actual Parts of the Calculator)



1. LCD Display



- Type : 7-segment LCD
- Digits : 12 digits
- Operating Voltage : 3V (typ.)
- Current Consumption : ~1–5 µA (standby)
- Temperature Range : -10°C to 60°C

2. Calculator IC (Controller)



- Type : Custom ASIC / LCD driver
- Supply Voltage : 3V (from battery/ solar cell)
- Operating Current : ~10–100 µA (typ.)
- Functions : Arithmetic logic, display control, power management

3. Solar Cell



- Type : Amorphous Silicon
- Output Voltage : ~1.5 – 2.0 V
- Output Current : ~1–10 mA (in bright light)
- Power : ~1–20 mW (typ.)
- Size (approx.) : 40 mm × 20 mm

4. Coin Cell Battery



- Type : Lithium (Li-MnO_2)
- Model : CR2032
- Nominal Voltage : 3V
- Capacity : 200–240 mAh
- Operating Temp. : -20°C to 60°C
- Shelf Life : ~5–10 years

5. PCB (Printed Circuit Board)



- Material : FR-4 (Glass Epoxy)
- Layers : Single sided
- Thickness : ~1.0 – 1.6 mm
- Operating Voltage : 3V
- Finish : HASL (Lead Free)
- Dimensions (approx.) : 140 mm × 100 mm (varies)

6. Conductive Rubber (Zebra Connector)



- Type : Zebra strip
- Material : Conductive elastomer
- Resistance : < 50 Ω (typ.)
- Operating Voltage : 3V
- Temperature Range : -20°C to 70°C

7. Silicone Rubber Keypad Sheet



- Material : Silicone rubber
- Hardness : 40–60 Shore A
- Operating Force : 100–300 g (typ.)
- Life Cycle : > 1 million presses
- Temperature Range : -20°C to 70°C

8. Key Caps (Plastic)



- Material : ABS Plastic
- Printing : Laser / Pad printing
- Size : ~8–12 mm (varies)
- Hardness : 70–100 Shore D
- Color : Black / Pink (function keys)

9. Resistors



- Type : Carbon Film / Metal Film
- Resistance Range : 10 Ω – 1 MΩ
- Power Rating : 1/8 W – 1/4 W
- Tolerance : ±1% – ±5%
- Operating Voltage : 50 V – 250 V

10. Capacitors



- Type : Electrolytic / Ceramic
- Capacitance : 10 µF – 100 µF
- Voltage Rating : 6.3 V – 25 V
- Tolerance : ±10% – ±20%
- Operating Temp. : -40°C to 85°C

11. Diodes



- Type : 1N4148 (Signal Diode)
- Max Reverse Voltage : 75 V
- Forward Current : 200 mA
- Forward Voltage : ~0.7 V
- Purpose : Protection / Logic

12. Transistors



- Type : BC547 / BC557 (NPN/PNP)
- Max Collector Voltage : 45 V
- Max Collector Current : 100 mA
- Gain (hFE) : 110 – 800
- Operating Temp. : -55°C to 150°C

13. Solar / Battery Wiring



- Type : Copper wire (insulated)
- Gauge : ~26–30 AWG
- Voltage Rating : 3V – 5V
- Current Capacity : 0.5 – 2 A
- Insulation : PVC

14. Screws & Fasteners



- Type : Self-tapping / Machine screw
- Material : Steel
- Size : M2 – M2.5 (approx.)
- Finish : Black / Zinc plated
- Purpose : Hold casing & PCB

15. Rubber Feet / Pads



- Material : Rubber (silicone/EPDM)
- Diameter : 6–10 mm (approx.)
- Thickness : 1–3 mm
- Hardness : 50–70 Shore A
- Purpose : Prevent slipping, absorb shock

Power Supply Specification

Solar Cell
1.5 – 2.0 V | 1–10 mA
(works in bright light)

+

Coin Cell Battery
3 V (CR2032)
200–240 mAh

Total Power System

- ✓ Solar cell provides main power in light.
- ✓ Battery provides backup when light is low.
- ✓ Both supply 3V to the circuit.

General Specifications (CASIO MJ-120D)

Display	12 digits (LCD)
Power Source	Solar + CR2032 battery
Operating Voltage	3V
Power Consumption	~1–5 µA (standby)
Operating Temperature	-10°C to 60°C
Dimensions (approx.)	149 × 106 × 30 mm
Weight (approx.)	~120 g

Key Features

- ✓ 12-digit large LCD display
- ✓ Two-way power (solar + battery)
- ✓ Auto power off
- ✓ Plastic keys with soft touch
- ✓ Durable and lightweight design

CASIO MJ-120D
Reliable • Efficient • Long Lasting

CASIO | MJ-120D (12 Digit) Calculator

Bill of Materials (BOM)

S. No.	Component / Part	Image / Example	Material	Quantity (Approx.)	Specification / Details	Function
1	Plastic outer casing		ABS Plastic	1 set	Molded plastic, front & back cover	Protects and holds all components
2	Printed Circuit Board (PCB)		FR-4 (Fiberglass)	1	Single sided PCB	Main electronic circuit
3	Calculator IC / Controller IC		Silicon	1	Custom IC (calculator logic)	Performs calculations and controls the calculator
4	LCD Display		LCD (Liquid Crystal)	1	12 digits, segment type	Displays numbers and symbols
5	LCD Conductive Rubber / Zebra Connector		Conductive Rubber	1	Zebra connector strip	Provides electrical connection between LCD and PCB
6	Silicone / Rubber Keypad Sheet		Silicone Rubber	1	Key matrix sheet	Transfers key presses to PCB
7	Individual Plastic Key Caps		ABS Plastic	~30–35	Various shapes and markings	User input for numbers and functions
8	Solar Cell		Amorphous Silicon	1	Small solar panel	Converts light into electrical energy
9	Coin-cell Battery		Lithium ($Li-MnO_2$)	1	CR2032 (3V)	Provides backup/operating power
10	Battery Contacts / Terminals		Stainless Steel	1 set	Spring type contacts	Connects battery to PCB
11	Diodes		Silicon	Several	1N4148 / similar	Rectification and circuit protection
12	Resistors		Carbon Film / Metal Film	Several	Various values (e.g., 10Ω, 1kΩ)	Current limiting and voltage control
13	Capacitors		Electrolytic / Ceramic	Several	e.g., 10µF, 100nF, 16V	Filtering and voltage stabilization
14	Transistors		Silicon	Several	e.g., BC547 / similar	Switching / control functions
15	PCB Tracks		Copper (on FR-4)	—	Printed tracks	Electrical connections between components
16	Connecting Wires		Copper Wire (insulated)	2–3	Thin insulated wires	Electrical connections, including solar cell
17	Screws		Steel	Several	Self-tapping / machine screws	Hold casing and components together
18	Rubber Feet / Pads		Rubber	4	Adhesive type	Prevent slipping and protect the base

Main Components for Presentation: PCB + Calculator IC + LCD Display + Solar Cell + Coin Battery + Keypad + Conductive Rubber Connector + Resistors + Capacitors + Diodes + Transistors + Plastic Casing + Connecting Wires + Screws.

(Power System: Solar cell + coin-cell battery = Two-way power supply)

CASIO MJ-120D (12 Digit) Calculator Complete Working Flow

Two Way Power
Solar Cell + Coin Battery

1 Power Supply Flow



Solar Cell
Converts light energy into electrical energy (works in bright light).

Provides power to the circuit →



Coin Cell Battery
Provides backup power when there is no light.

Both sources feed the power management circuit →



Diodes + Resistors + Capacitors
Regulate and stabilize the voltage (about 3V) to the main circuit.

Stable DC power (3V) →



Main PCB & Calculator IC
Supplies power to all components and controls the operations.

2 Key Press → Logic → Display Flow

1. 

Key Pressed
You press a key (e.g., 5, +, = etc.). The key cap pushes the silicone rubber.

2. 

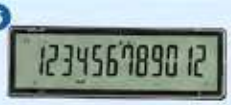
Silicone Rubber Sheet
Conductive pad closes the circuit at the specific key position on the PCB.

3. 

PCB Registers the Input
The circuit detects the key press and sends the signal to the calculator IC.

4. 

Calculator IC (Controller)
Processes the input, performs the calculation (addition, subtraction, etc.) and controls the display.

5. 

LCD Display
The IC sends the result to the LCD through the zebra connector. The liquid crystal changes state to show the numbers and symbols.

3 How Each Important Component Works

1. **Solar Cell**

Generates electricity from light.

2. **Coin Battery**

Provides power when there is no light.

3. **Diodes**

Allow current in one direction (protection & rectification).

4. **Resistors**

Limit current and control voltage.

4. **Capacitors**

Smooth and stabilize the voltage.

5. **Transistors**

Used for switching and control functions (in the IC circuit).

2. **Calculator IC**


Handles all calculations and controls the display, keys and power management.

3. **LCD Display**

Shows numbers and symbols using liquid crystals.

3. **Zebra Connector**

Transfers signals between PCB and LCD.

4. **Keypad (Rubber + Keys)**


Detects key presses and sends signals to PCB.

5. **PCB Tracks**

Connect all components electrically.

5. **Casing**

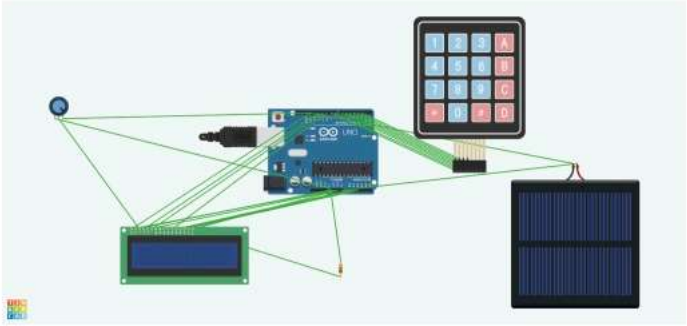
Holds and protects all parts.

4 Final Working Flow (Summary)

Light or Battery provides power → Voltage is regulated (diodes + resistors + capacitors) → Key press is detected (key → rubber → PCB) → IC processes and calculates → Result is sent to LCD display

In Short
The solar cell and battery give power, the PCB and IC handle the logic, the keypad gives input, and the LCD shows the result. All components work together to make the calculator function smoothly.

4:35
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